

Node Structure Visualization

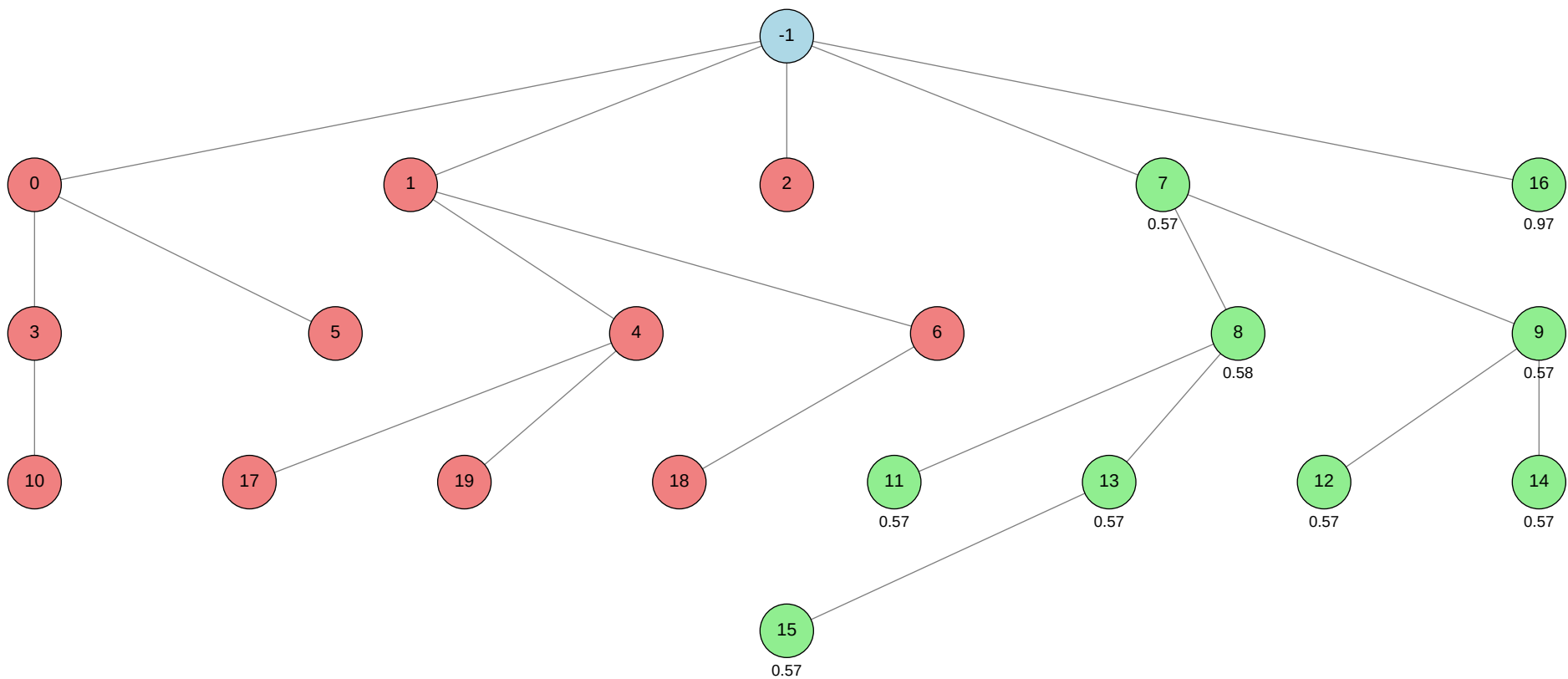
Click on any node in the tree to jump to its detailed information.

Total Nodes: 21 | Best Validation Score: 0.9652 (Node 16)

Node Tree:

Table of Contents:

Node ID	Stage	Status	Tool
<-1			
>0			autogluon.multimodal
>1			machine learning
>2			autogluon.tabular
>3			autogluon.multimodal
>4			machine learning
>5			autogluon.multimodal
>6			machine learning
>7			autogluon.tabular
>8			autogluon.tabular
>9			autogluon.tabular
>10			autogluon.multimodal
>11			autogluon.tabular
>12			autogluon.tabular
>13			autogluon.tabular
>14			autogluon.tabular
>15			autogluon.tabular
>16			autogluon.multimodal
>17			machine learning
>18			machine learning
>19			machine learning



Status	Color
Success	Green
Failure	Red
Neutral	Blue

Node Details:

Node -1 (root)

Property	Value
uct_value	-0.1717
ctime	2026-05-22 16:55:49
debug_attempts	0
depth	0
execution_time	0.0
expected_child_count	0
failure_visits	11
id	-1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
num_children	5
prev_tutorial_prompt	None
stage	root
time_step	-1
tool_used	
tools_available	[]
unvalidated_visits	0
validated_reward	5.54118
validated_visits	9
validation_score	None
visits	20
parent_id	None
child_ids	[0, 1, 2, 7, 16]

Node 0 (evolve)

Property	Value
uct_value	0.9737
ctime	2026-05-22 16:56:52
debug_attempts	2
depth	1
execution_time	0.0
expected_child_count	0
exploitation	-0.25
exploration	1.1772322201769845
failure_penalty	-0.25
failure_visits	4
id	0
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	2
num_children	2
prev_tutorial_prompt	None
stage	evolve
time_step	0
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	4
parent_id	-1
child_ids	[3, 5]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 47-addition_2/node_0/conda_env

Resolved 235 packages in 1.56s

Uninstalled 1 package in 1.91s

Installed 233 packages in 2m 23s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2
- + hyperopt==0.2.7

+ idna==3.16
... [truncated, 10002 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the training data contains a third label, 'non-neoplastic', in addition to the expected 'benign' and 'malignant' classes. The code attempts to compute class weights and configure focal loss for binary classification, but the presence of a third, unexpected string label causes a ValueError ("too many dimensions 'str'") when converting class weights to a torch tensor in AutoGluon.

SUGGESTED_FIX:

Before training, filter the training DataFrame to include only rows where the label is either 'benign' or 'malignant'. Exclude all samples with the label 'non-neoplastic' from both training and validation splits, so that only the two intended classes are present when computing class weights and fitting the model.

Node 1 (evolve)

Property	Value
uct_value	0.6658
ctime	2026-05-22 17:07:27
debug_attempts	2
depth	1
execution_time	0.0
expected_child_count	0
exploitation	-0.3
exploration	0.813817360151413
failure_penalty	-0.3
failure_visits	6
id	1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	3
num_children	2
prev_tutorial_prompt	None
stage	evolve
time_step	1
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	6
parent_id	-1
child_ids	[4, 6]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

warning: Requirements file `~/home/anri21/.conda/envs/mlzero\_env/lib/python3.11/site-packages/autogluon/assistant/tools\_registry/machine learning/requirements.txt` does not contain any dependencies

Using Python 3.11.15 environment at: 47-addition_2/node_1/conda_env

Resolved 79 packages in 448ms

Uninstalled 1 package in 1.74s

Installed 78 packages in 40.56s

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiosignal==1.4.0

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ anyio==4.13.0

+ attrs==26.1.0

+ certifi==2026.5.20

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ click==8.4.1

+ cuda-bindings==13.2.0

+ cuda-pathfinder==1.5.4

+ cuda-toolkit==13.0.2

+ datasets==4.8.5

+ dill==0.4.1

+ filelock==3.29.0

+ frozenlist==1.8.0

+ fsspec==2026.2.0

+ h11==0.16.0

+ hf-xet==1.5.0

+ httpcore==1.0.9

+ httpx==0.28.1

+ huggingface-hub==1.16.1

+ idna==3.16

+ jinja2==3.1.6

```
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ s... [truncated, 7667 chars total]
```

Error Analysis:

ERROR_SUMMARY: TensorFlow failed to initialize the cuDNN DNN library on the GPU, resulting in a runtime error during feature extraction with EfficientNetB0. The error "Could not create cudnn handle: CUDNN_STATUS_NOT_INITIALIZED" suggests a mismatch or incompatibility between the installed CUDA/cuDNN libraries and the NVIDIA GPU driver, or an incomplete/corrupted CUDA/cuDNN installation. As a result, no model was trained or predictions produced.

SUGGESTED_FIX:

Check that your NVIDIA GPU driver version is compatible with the installed CUDA and cuDNN versions (driver 565.57.1 is reported; check TensorFlow's compatibility table for required versions). Ensure that the CUDA and cuDNN libraries are correctly installed and accessible in your environment, and that the correct versions are being used by TensorFlow. If necessary, update or reinstall the GPU driver, CUDA, and cuDNN to versions known to work together with your TensorFlow version, then restart the environment and rerun the script.

Node 2 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-22 17:16:56
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	0.986235895764877
failure_penalty	-0.0
failure_visits	1
id	2
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	2
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 47-addition_2/node_2/conda_env

Resolved 239 packages in 2.02s

Uninstalled 1 package in 1.86s

Installed 237 packages in 1m 44s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + daal==2025.9.0
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2

+ hyperopt==0.... [truncated, 174846 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the test set provided to ``predictor.predict_proba()`` only contains the column ``image_name``, while the trained AutoGluon Tabular model expects the columns ``skin_tone`` and ``alternative_skin_tone`` (used as features during training). As a result, a `KeyError` is raised when the model attempts to access these missing columns in the test data.

SUGGESTED_FIX:

Modify your test set preprocessing so that the `DataFrame` passed to ``predictor.predict_proba()`` includes all feature columns used during training (``skin_tone``, ``alternative_skin_tone``, and ``image_name``). If the test set does not have these columns, you must infer or impute reasonable values for them (e.g., set to a default or use predictions from another model), or adjust your training to only use features available at test time. Ensure the columns and their order match those used in training.

Node 3 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-22 17:26:43
debug_attempts	2
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.1772322201769845
failure_penalty	-0.0
failure_visits	2
id	3
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	3
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	2
parent_id	0
child_ids	[10]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 47-addition_2/node_3/conda_env

Resolved 235 packages in 2.86s

Uninstalled 1 package in 1.86s

Installed 233 packages in 1m 35s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0

```
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.16
... [truncated, 9772 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs because the class weights (``alpha``) for focal loss are being passed as a dictionary mapping string labels to weights, but AutoGluon expects ``alpha`` to be a list or numpy array of floats (not a dict or list of strings), with the order matching the internal class index order. When the code tries to convert the dict to a tensor, it encounters a string key, resulting in the ValueError: "too many dimensions 'str'".

SUGGESTED_FIX:

Modify the code so that ``alpha`` (class weights) is passed as a list of floats, ordered to match the class indices used by AutoGluon (typically sorted label order or as inferred by the model). Do not use a dictionary or include string labels; instead, ensure ``class_weights_list = [weight_for_class0, weight_for_class1]`` where the order matches the label encoding (e.g., ``['benign', 'malignant']``). Pass this list directly as the value for ``"optim.focal_loss.alpha"`` in the hyperparameters dictionary.

Node 4 (debug)

Property	Value
uct_value	0.9261
ctime	2026-05-22 17:34:40
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	3
id	4
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	2
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug
time_step	4
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	3
parent_id	1
child_ids	[17, 19]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 47-addition_2/node_4/conda_env
Resolved 79 packages in 393ms
Uninstalled 1 package in 1.71s
Installed 78 packages in 38.15s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```

- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.16.1
- + idna==3.16
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.14.0a1
- + pydantic-core==2.47.0
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+ s... [truncated, 7563 chars total]

Error Analysis:

ERROR_SUMMARY: TensorFlow failed to initialize the cuDNN DNN library on the GPU due to a "CUDNN_STATUS_NOT_INITIALIZED" error, which is typically caused by an incompatible or insufficient NVIDIA GPU driver version relative to the installed CUDA/cuDNN libraries. As a result, feature extraction using EfficientNetB0 could not proceed, leading to a runtime error and script termination.

SUGGESTED_FIX: Check that your NVIDIA GPU driver version is compatible with the installed CUDA and cuDNN versions required by your TensorFlow installation. Update your GPU driver to the latest supported version, and ensure that the CUDA and cuDNN libraries match the versions expected by TensorFlow; then restart your environment and rerun the script.

Node 5 (debug)

Property	Value
uct_value	1.6649
ctime	2026-05-22 17:43:41
debug_attempts	2
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	1
id	5
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	5
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	1
parent_id	0
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 47-addition_2/node_5/conda_env

Resolved 235 packages in 1.38s

Uninstalled 1 package in 1.70s

Installed 233 packages in 1m 38s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0

```
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.16
... [truncated, 9849 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs because the class weights (``alpha``) passed to AutoGluon's FocalLoss are a list of strings instead of floats. This happens because the code constructs ``class_weights_list`` using a dictionary with string keys (the class names), but the resulting list may contain string representations or be misaligned with the expected numeric format. FocalLoss expects ``alpha`` to be a list or array of floats, not strings, and thus raises a ``ValueError: too many dimensions 'str'`` when trying to convert the list to a PyTorch tensor.

SUGGESTED_FIX:

Check how ``class_weights_list`` is constructed and ensure it is a list of floats (not strings). Explicitly convert each value in ``class_weights_list`` to a float before passing it to the hyperparameters (e.g., use ``[float(class_weights[lab]) for lab in label_order]``). Double-check that the order matches the label encoding used by AutoGluon, and that the list contains only numeric values.

Node 6 (debug)

Property	Value
uct_value	1.3384
ctime	2026-05-22 17:51:37
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	2
id	6
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have then
stage	debug
time_step	6
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	2
parent_id	1
child_ids	[18]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 47-addition_2/node_6/conda_env
Resolved 79 packages in 357ms
Uninstalled 1 package in 1.78s
Installed 78 packages in 38.16s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```

- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.16.1
- + idna==3.16
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.14.0a1
- + pydantic-core==2.47.0
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+ s... [truncated, 7652 chars total]

Error Analysis:

ERROR_SUMMARY: TensorFlow failed to initialize the cuDNN DNN library on the GPU, resulting in a runtime error during feature extraction with EfficientNetB0. The error log shows "Could not create cudnn handle: CUDNN_STATUS_NOT_INITIALIZED" and "DNN library initialization failed," which is typically caused by a mismatch between the installed CUDA/cuDNN libraries and the NVIDIA driver version, or by insufficient available GPU memory/resources.

SUGGESTED_FIX: Check that your NVIDIA driver version is fully compatible with the installed CUDA and cuDNN versions (driver 565.57.1 is reported, but the runtime/toolkit/DNN versions may not match). Ensure that no other processes are occupying GPU memory, and consider reducing the batch size in feature extraction to lower memory usage. If the problem persists, try reinstalling TensorFlow with versions specifically built for your CUDA/cuDNN stack, or run the script on CPU as a fallback.

Node 7 (debug)

Property	Value
uct_value	0.8735
avg_raw_score	0.57199625
ctime	2026-05-22 18:01:06
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.008213105657426693
exploration	0.4289194095773362
failure_penalty	-0.0
failure_visits	0
id	7
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.008213105657426693
num_children	2
prev_tutorial_prompt	None
stage	debug
time_step	7
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.008213105657426693
validated_reward	4.57597

validated_visits	8
validated_weight	1.0
validation_score	0.5718
visits	8
parent_id	-1
child_ids	[8, 9]

Node 8 (evolve)

Property	Value
uct_value	1.0286
avg_raw_score	0.5735466666666666
ctime	2026-05-22 18:08:27
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.6529152915291482
exploration	1.138807119808647
failure_penalty	-0.0
failure_visits	0
id	8
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.6529152915291482
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinds dataset.
stage	evolve
time_step	8
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.6529152915291482
validated_reward	2.28938
validated_visits	4
validated_weight	1.0
validation_score	0.5756
visits	4
parent_id	7
child_ids	[11, 13]

Node 9 (evolve)

Property	Value
uct_value	1.1844
avg_raw_score	0.5715933333333333
ctime	2026-05-22 18:19:09
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.33058305830581863
exploration	1.138807119808647
failure_penalty	-0.0
failure_visits	0
id	9
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.33058305830581863
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinds dataset.
stage	evolve
time_step	9
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.33058305830581863
validated_reward	1.71478
validated_visits	3
validated_weight	1.0
validation_score	0.5728
visits	3
parent_id	7
child_ids	[12, 14]

Node 10 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-22 18:29:57
debug_attempts	2
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	10
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	10
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	3
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 47-addition_2/node_10/conda_env

Resolved 235 packages in 3.05s

Uninstalled 1 package in 1.89s

Installed 233 packages in 1m 52s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.1

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.16... [truncated, 17114 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs during inference when calling ``predictor.predict_proba(test_pred_df)``. The ValueError indicates that the test dataframe only contains the columns ``['image_name', 'image']``, but is missing the required column ``['skin_tone']``, which was present during training. AutoGluon MultiModal expects all

columns used during training (except the label) to also be present at inference time.

SUGGESTED_FIX:

Modify your test dataframe before prediction to include a `skin\_tone` column. If true skin tone values are unavailable for the test set, add a placeholder column (e.g., fill with a default or median value, or use a neutral value such as -1 or the most common skin tone from the training set). Ensure the test dataframe passed to `predict\_proba` has columns `['image\_name', 'image', 'skin\_tone']` in addition to any required columns, matching the training data structure (excluding the label).

Node 11 (evolve)

Property	Value
uct_value	1.6695
avg_raw_score	0.5706
ctime	2026-05-22 19:30:45
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.16666666666666055
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	0
id	11
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.16666666666666055
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinds dataset.
stage	evolve
time_step	11
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.166666666666666055
validated_reward	0.5706
validated_visits	1
validated_weight	1.0
validation_score	0.5706
visits	1
parent_id	8
child_ids	[]

Node 12 (evolve)

Property	Value
uct_value	1.4913
ctime	2026-05-22 19:41:52
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
failure_visits	0
id	12
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinder dataset.
stage	evolve
time_step	12
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.5724
validated_visits	1
validation_score	0.5724
visits	1
parent_id	9
child_ids	[]

Node 13 (evolve)

Property	Value
uct_value	1.1844
avg_raw_score	0.57439
ctime	2026-05-22 19:52:05
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.7920792079207841
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	0
id	13
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.7920792079207841
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinds dataset.
stage	evolve
time_step	13
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.7920792079207841
validated_reward	1.14313
validated_visits	2
validated_weight	1.0
validation_score	0.5744
visits	2
parent_id	8
child_ids	[15]

Node 14 (evolve)

Property	Value
uct_value	1.4842
ctime	2026-05-22 20:03:15
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
failure_visits	0
id	14
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinder dataset.
stage	evolve
time_step	14
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.56959
validated_visits	1
validation_score	0.5696
visits	1
parent_id	9
child_ids	[]

Node 15 (evolve)

Property	Value
uct_value	1.1772
ctime	2026-05-22 20:14:19
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	15
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinder dataset.
stage	evolve
time_step	15
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.56874
validated_visits	1
validation_score	0.5687
visits	1
parent_id	13
child_ids	[]

Node 16 (debug)

Property	Value
uct_value	3.4474
ctime	2026-05-22 20:24:53
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	0
id	16
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✓
num_children	0
prev_tutorial_prompt	None
stage	debug
time_step	16
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96521
validated_visits	1
validation_score	0.9652
visits	1
parent_id	-1
child_ids	[]

Node 17 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-22 21:25:38
debug_attempts	2
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	17
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug
time_step	17
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 47-addition_2/node_17/conda_env
Resolved 79 packages in 431ms
Uninstalled 1 package in 1.67s
Installed 78 packages in 46.91s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.16

```

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ ... [truncated, 7720 chars total]
```

Error Analysis:

ERROR_SUMMARY: TensorFlow failed to initialize the cuDNN DNN library on the GPU, resulting in a runtime error during feature extraction with EfficientNetB0. The error log shows "Could not create cudnn handle: CUDNN_STATUS_NOT_INITIALIZED" and suggests a possible mismatch or insufficiency in the NVIDIA driver, CUDA, or cuDNN versions. This prevented any features from being extracted and caused the script to crash before model training.

SUGGESTED_FIX: Verify that the installed NVIDIA driver, CUDA toolkit, and cuDNN versions are fully compatible with each other and with the TensorFlow version in your environment. Check that the driver version is recent enough for your GPU (A100 requires a new driver), and ensure that the CUDA/cuDNN libraries are correctly installed and visible to TensorFlow (e.g., via LD_LIBRARY_PATH). If running in a container or cluster, confirm that GPU access is enabled and not restricted by resource limits or environment misconfiguration.

Node 18 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-22 21:35:48
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	18
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug
time_step	18
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	6
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 47-addition_2/node_18/conda_env
Resolved 79 packages in 1.17s
Uninstalled 1 package in 1.70s
Installed 78 packages in 43.89s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.16

```

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ ... [truncated, 7722 chars total]
```

Error Analysis:

ERROR_SUMMARY: TensorFlow failed to initialize the cuDNN DNN library on the GPU, resulting in a runtime error during feature extraction with EfficientNetB0. The error "Could not create cudnn handle: CUDNN_STATUS_NOT_INITIALIZED" suggests a mismatch between the installed CUDA/cuDNN libraries and the NVIDIA driver, or insufficient available GPU memory/resources at runtime. This prevented any features from being extracted and caused the pipeline to crash before model

training.

SUGGESTED_FIX:

Check that your NVIDIA driver version is fully compatible with the installed CUDA and cuDNN versions (driver $\geq$ CUDA runtime/toolkit version, and cuDNN matches CUDA). Ensure no other processes are occupying GPU memory, and consider rebooting or freeing up GPU resources. If the problem persists, try downgrading or upgrading TensorFlow, CUDA, or cuDNN to ensure all versions are mutually compatible, and verify that the correct libraries are being loaded by TensorFlow at runtime.

Node 19 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-22 21:45:05
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	19
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	<pre>### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them</pre>
stage	debug
time_step	19
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 47-addition_2/node_19/conda_env
Resolved 79 packages in 373ms
Uninstalled 1 package in 1.71s
Installed 78 packages in 37.98s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.16

```

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ ... [truncated, 7661 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs during feature extraction with TensorFlow/Keras EfficientNetB0 because the cuDNN DNN library fails to initialize on the GPU. The logs indicate a possible CUDA/cuDNN driver mismatch or insufficient GPU resources, resulting in TensorFlow being unable to create a cuDNN handle and halting execution with a "DNN library initialization failed" error.

SUGGESTED_FIX: Check that the installed CUDA and cuDNN versions are fully compatible with both your NVIDIA driver and the TensorFlow version in your environment. Ensure that the NVIDIA driver is up to date, that no other processes are exhausting GPU memory, and that the correct CUDA/cuDNN libraries are visible in your environment (e.g., via LD_LIBRARY_PATH). If compatibility issues persist, consider forcing TensorFlow to run on CPU by setting the environment variable CUDA_VISIBLE_DEVICES="" before running the script to bypass GPU initialization errors.